Industry-Standard Logistics & Transportation KPI Formulas

These are the globally recognized KPI metrics used by major logistics players (DHL, FedEx, UPS, and top 3PLs). These should be pre-load as the "Default Templates" in the configurable page.

1. **On-Time In-Full (OTIF) Rate**

- This is the most critical metric in modern logistics, heavily enforced by major retailers and used by global couriers like DHL and FedEx
    - **Formula:** (Number of Orders Delivered On-Time AND In-Full / Total Number of Orders) × 100
    - **Industry Benchmark:** 95% to 98% is considered good; anything below 95% incurs penalties in B2B logistics
    - **STARS Application:** Measures if the Encoder/Courier completed the task within the SLA deadline (On-Time) and with all required documents/attachments (In-Full).

2. **Order Accuracy Rate (Perfect Order Rate)**

- Measures the percentage of orders processed without any errors (no push-backs, no missing docs, no wrong routing). Widely used by 3PLs and warehousing operations
    - **Formula:** (Number of "Perfect" Orders / Total Number of Orders Fulfilled) × 100 (Note: A "Perfect" order in STARS = Completed on the first try with ZERO "Push Backs" from the Coordinator).
    - **Industry Benchmark:** 96% to 98% is strong; above 98% is exceptional in the 3PL/courier industry

3. **Task Completion Rate (Internal Operational KPI)**

- Measures workforce productivity and capacity utilization.
    - **Formula:** (Total Tasks Marked "Completed" by Employee / Total Tasks Assigned to Employee) × 100
    - **Industry Benchmark:** Varies, but top-performing logistics teams aim for >90% completion rate within the assigned

4. **Average Handling Time (AHT) / Cycle Time**

- Measures the average time taken to move a task from "Not Started" to "Completed".
    - **Formula:** Total Time Spent on All Tasks / Total Number of Tasks Completed
    - **Industry Benchmark:** Depends on task type, but reducing AHT by 10-15% year-over-year is a standard logistics efficiency goal

MUST Be Inside the "Configurable KPI Page" (Manager View)

A. The KPI Library & Selector

- **Pre-built Templates:** A dropdown or card grid showing the standard formulas above (OTIF, Accuracy, etc.).
- **Custom Formula Builder:** A simple UI where the Manager can create a new KPI (e.g., (Field A / Field B) * 100).
- **Weighting System (Crucial for Performance Evaluation):**
  - Managers must be able to assign a weight (%) to each KPI to calculate a final "Employee Performance Score".
  - *Example: OTIF (40%) + Order Accuracy (40%) + Task Completion (20%) = 100% Total Performance Score.*

B. Threshold & Target Configuration (Traffic Light System)

- Managers need to define what "Good" and "Bad" look like for their specific operations.
  - **Target (Green):** e.g., ≥ 98%
  - **Warning (Yellow/Amber):** e.g., 95% – 97.9%
  - **Critical (Red):** e.g., < 95%
  - **Why this matters:** This directly feeds into the Biomarker AI, allowing the AI to flag employees who fall into the "Red" zone.

C. Scope & Filtering Controls

- **Timeframe Selector:** Daily, Weekly, Monthly, Quarterly, Year-to-Date (YTD).
- **Role/Department Filter:** Apply this specific KPI configuration to "Couriers" only, or "Encoders" only. (A courier's KPI should not be judged the same way as an encoder's).
- **Employee Selector:** Drill down from Department → Team → Specific Employee.

D. Data Source Mapping (Backend Requirement)

- For the system to calculate the formula, the Manager must map the formula variables to actual database tables.

E. The Performance Evaluation Output

- **Individual Scorecard:** A clean view showing the employee's name, their final weighted score (e.g., "94.5%"), and a breakdown of how they scored on each individual KPI.
- **Export Functionality:** Button to export the evaluation as a PDF or Excel file for HR records.

Technical & Business Considerations

1. **Historical Data Recalculation:** If a Manager changes the KPI weights today, does it recalculate past months' evaluations?
    - No. Configurable KPIs should only apply prospectively (from the date of saving onward), or the system must clearly state "Recalculating historical data..." to avoid HR disputes.

2. **Data Integrity:** The KPI calculator must pull from immutable data (e.g., the Audit Log or Completed Tasks table), not from editable fields, to prevent employees from "gaming" their performance scores.

3. **Role Separation:** Only the Manager should have access to the Configuration page. Operation Employees should only see a read-only version of their own KPI dashboard (no weighting or formula editing).

References:

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